MTH1001

UNIVERSITY OF EXETER

COLLEGE OF ENGINEERING, MATHEMATICS AND PHYSICAL SCIENCES

MATHEMATICS

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Mathematical Structures

Module Leader: Prof. Nigel Byott

Other Examiner: Dr Gihan Marasingha

Duration: 2 HOURS (+30 minutes upload time)

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.

This is an **OPEN NOTE** examination.

SECTION A

1. (a) In this question, select **one or more** possible deductions from a list of premises. You need not provide reasoning.

   (i) Given the premises $P \to Q$ and P, which of the following can be deduced: P, Q, $P \wedge Q$, $P \vee Q$, $\bot$?

   (ii) Given the premises $P \to Q$ and Q, which of the following can be deduced: P, Q, $P \wedge Q$, $P \vee Q$, $\bot$?

   (iii) Given the premises $P \vee \neg P$ and $Q \vee \neg Q$, which of the following can be deduced: P, Q, $P \wedge Q$, $P \vee Q$, $\bot$?

   (iv) Given the premise $P \wedge \neg P$, which of the following can be deduced: P, Q, $P \wedge Q$, $P \vee Q$, $\bot$? (8)

   (b) (i) Let A and B be types. Let $f : A \to B$ be a function. Give the definition of 'f is injective'.

   (ii) Which of the following functions are injective? In each case give a proof of your answer. Here, $\mathbb{N}$ is the type of non-negative integers and $\mathbb{Z}$ is the type of integers.

   $f : \mathbb{N} \to \mathbb{Z}$, where $f(x) = (x-1)^2$,
   $g : \mathbb{N} \to \mathbb{Z}$, where $g(y) = (y+1)^2$. (10)

   (c) Let $A = \{1, 2, 3, 4\}$, $B = \{2, 4, 6\}$, $C = \{1, 6, 8\}$, $D = \{7, 9\}$ be sets on the type $\mathbb{N}$. Write each of the following sets as lists of elements enclosed in braces. You need not provide reasoning.

(i) $A \cup B$,	(ii) $A \cap C$,	(iii) $C \setminus A$,
(iv) $\mathcal{P}(D)$,	(v) $B \times D$,	(vi) $\mathcal{P}(B \cap D)$.

   (12)

   (d) Let $z = 4 + i$ and $w = 3 - i$ be complex numbers, where $i^2 = -1$.

   (i) Write each of $z + w$, $z - w$, and zw in the form $x + iy$ for real numbers x and y.

   (ii) Find all complex cube roots of $27\sqrt{2}(1 - i)$. Express your answers in polar form, ensuring you give the principal value of each argument. (10)

   (e) (i) Let $(a_n)_{n=0}^{\infty}$ be a sequence of real numbers. Let a be a real number. Define what it means for $(a_n)_{n=0}^{\infty}$ to converge to a.

   (ii) Let $(a_n)_{n=0}^{\infty}$ and $(b_n)_{n=0}^{\infty}$ be sequences of real numbers. Let a and b be real numbers. Given that $(a_n)_{n=0}^{\infty}$ and $(b_n)_{n=0}^{\infty}$ converge to a and b, respectively, prove that the sequence $(a_n + b_n)_{n=0}^{\infty}$ converges to $a + b$. (10)

[**50**]

SECTION B

2. (a) Let P, Q, and R be propositions. Given $h_1 : P \leftrightarrow Q \vee R$, $h_2 : P$, and $h_3 : \neg Q$, prove R. Your proof should only use the rules of inference of the propositional logic and not depend on classical reasoning. (12)

   (b) Let T be a set of real numbers. For a real number u to be a smallest element of T means $(u \in T) \wedge (\forall x \in T, u \leq x)$.

   (i) Prove that the set $S = \{x : \mathbb{R} \mid 1 < x\}$ has no smallest element.

   (ii) Consider the statement, 'for every set T of real numbers, if T has a lower bound and T is non-empty, then T has a smallest element'. Give a brief proof of either this statement or its negation. (13)

   [25]

3. (a) Let A and B be types and let $f : A \to B$ be a function.

   (i) Define what it means for f to be *surjective*. Give the definition of a *right inverse* of f.

   (ii) Prove that if f has a right inverse, then f is surjective. (12)

   (b) A binary operator $\oplus$ is defined on $\mathbb{N}$ recursively by the rules:

   - for every $x : \mathbb{N}$, $x \oplus 0 = S(x)$,
   - for all $x, y : \mathbb{N}$, $x \oplus S(y) = S(S(x \oplus y))$.

   Prove for all $x, y : \mathbb{N}$, that $x \oplus y = (x + 1) + 2y$. You may use all familiar properties of natural number arithmetic, proof by mathematical induction, and the fact that for every $\alpha : \mathbb{N}$, $S(\alpha) = \alpha + 1$. (13)

   [25]

4. (a) Prove or disprove the following statement.

   For every sequence $(a_n)_{n=0}^{\infty}$ of real numbers, for every real number a, if $\sum_{n=0}^{\infty} a_n$ converges with limit a, then $\sum_{n=0}^{\infty} a_n^2$ converges with limit a^2. (12)

   (b) (i) Let A and B be types. Let $f : A \to B$ be a function. Let S be a set on A. Give the definition of $f(S)$.

   (ii) Further, let C be a type and let $g : B \to C$ be a function. Prove that $g(f(S)) \subseteq (g \circ f)(S)$. (13)

   [25]